

Kuan-Cheng (James) Ku

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Education

- National Taiwan University** *Bachelor in Information Management* 2021 – 2025
- Overall GPA: 3.88, CS-related Elective GPA: 4.11; Key Courses: Web Programming, Applied Deep Learning, Introduction to Information Retrieval and Text Mining, Cloud Native Application Development, Systems Analysis and Design
 - Teaching Assistant: Assisted in preparing materials & exams and hosted TA sessions, serving 500+ students across 5 courses, including Web Programming, Data Structure, Systems Analysis & Design, Calculus and Marketing Management.
 - Lecturer: Worked for Information Clubs of Taipei Municipal Zhongshan Girls & Datong High School and [Sprout from NTU CSIE](#), creating materials and delivering lectures in C++ & Python Programming for over 3 years.

Experience

- Research Development Intern, Microsoft** *TSMC OnePlatform Team, ISD* Apr 2025 – Present
- Led unit & end-to-end testing initiatives by creating and conducting comprehensive test plans consisting of 400+ test cases, and designed & hosted product training sessions to educate key users.
 - Opened and assisted in resolving 150+ bugs in ADO, implemented POC experiments for frontend bugs, and organized interlock meetings between MS, ASUS and ADV to facilitate collaboration among engineers from three vendors.
- Dev Lead, NTU Art Festival 2024** *Riddle City Treasure Hunt Team* 🌐 Website Jan 2024 – Jun 2024
- Architected and led the development of an online city treasure hunt gaming website with Next.js integrating physical riddles on campus, creating immersive experience that gained 300+ active users during the event lasting 1 week.

Achievements

- TSMC IT CareerHack 2024: 2nd Place** 🏆 📄 deVopSCoDe Feb 2025
- Developed an AI maintenance toolkit, including web and VSCode interfaces, using Vertex AI, FastAPI, LangChain, and deployed on GCE and GKE within two days. Selected as one of 24 finalists from 168 teams and achieved second place.
- Taiwan Digital Innovation Awards 2025: Honorable Mention** 🏆 📄 RenderResume Jun 2025 - Oct 2025
- Developed an AI resume editor featuring evaluation, STAR restructuring, and intelligent Q&A optimization using OpenAI API, Next.js, and Supabase. Selected as one of 14 finalists from 100+ teams and received Honorable Mention (Top 7).

Projects

- One-Click Outfit** 🌐 (Bachelor Project) 📄 rmmfj/outfit-recommendation-app Jan 2024 – Jan 2025
- Architected and led fullstack development of an AI-driven platform that provides personalized fashion recommendations with Next.js and Flask, achieving a successful product launch with 300+ users within 2 weeks.
 - Deployed a scalable system architecture with Docker on a VPS, and reduced processing time by 70% by migrating no-sql database to PostgreSQL and installing vector indexing.
- NCHU Tutoring Program Dashboard (Freelance Project)** Apr 2025 - Sep 2025
- Architected and developed, as a solo freelancer, a full-stack student management platform using Next.js and Supabase to serve 300+ MAU, integrating task management, dynamic form builders, real-time analytics dashboard, and email notification system with Resend, file management with Google Drive API, reducing administrative overhead.
 - Designed a comprehensive RBAC security architecture, maintaining complete data isolation and compliance.
- StreamBench Agent: Continuous Improvement of LLM (Course Project)** Oct 2024 – Dec 2024
- Engineered NeurIPS 2024 StreamBench-inspired prompt optimization pipeline for medical diagnosis, boosting accuracy from 50.17% to 73.64% via clinical framing, structured reasoning, and negative example learning.

Leadership and Activities

- Director, NTU Information Management Student Association** *IT Department* Jun 2022 – Jan 2024
- Co-founder, NTUIM Competitive Programming Team** Jan 2022 – Jun 2023
- Chief Organizer, 2023 NTUIM Programming Contest: PDAO** Dec 2022 – May 2023
- Deputy Organizer, 2024 NTUIM Camp: Information Fog** Jun 2023 – Jan 2024

Skills and Languages

- Languages:** Python, C/C++, HTML5, CSS3, JavaScript, TypeScript, SQL **Chinese:** Native
- Frameworks:** Express, React, Next, Tailwind, LangChain, Pytorch, Flask, FastAPI **English:** Fluent
- Tools:** Git/GitHub, Docker, GCP (GCE, GKE, Vertex AI), Vercel, Azure DevOps TOEIC 950